

Project 2, parts 4 and 5: RNA-seq analysis

2025-05-08

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(edgeR)
```

```
## Loading required package: limma
```

```
library(limma)
library(Glimma)
library(ggplot2)
library(RColorBrewer)
library(DESeq2)
```

```
## Loading required package: S4Vectors
## Loading required package: stats4
## Loading required package: BiocGenerics
## Loading required package: generics
##
## Attaching package: 'generics'
##
## The following object is masked from 'package:lubridate':
##
##   as.difftime
##
## The following object is masked from 'package:dplyr':
##
##   explain
##
## The following objects are masked from 'package:base':
##
##   as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##   setequal, union
```

```

##
##
## Attaching package: 'BiocGenerics'
##
## The following object is masked from 'package:limma':
##
##     plotMA
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
##
## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##     unsplit, which.max, which.min
##
##
## Attaching package: 'S4Vectors'
##
## The following objects are masked from 'package:lubridate':
##
##     second, second<-
##
## The following objects are masked from 'package:dplyr':
##
##     first, rename
##
## The following object is masked from 'package:tidyr':
##
##     expand
##
## The following object is masked from 'package:utils':
##
##     findMatches
##
## The following objects are masked from 'package:base':
##
##     expand.grid, I, unname
##
## Loading required package: IRanges
##
## Attaching package: 'IRanges'
##
## The following object is masked from 'package:lubridate':
##

```

```

##      %within%
##
## The following objects are masked from 'package:dplyr':
##
##      collapse, desc, slice
##
## The following object is masked from 'package:purrr':
##
##      reduce
##
## The following object is masked from 'package:grDevices':
##
##      windows
##
## Loading required package: GenomicRanges
## Loading required package: Seqinfo
## Loading required package: SummarizedExperiment
## Loading required package: MatrixGenerics
## Loading required package: matrixStats
##
## Attaching package: 'matrixStats'
##
## The following object is masked from 'package:dplyr':
##
##      count
##
## Attaching package: 'MatrixGenerics'
##
## The following objects are masked from 'package:matrixStats':
##
##      colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
##      colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
##      colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
##      colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
##      colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
##      colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
##      colWeightedMeans, colWeightedMedians, colWeightedSds,
##      colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
##      rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
##      rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
##      rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
##      rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
##      rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
##      rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
##      rowWeightedSds, rowWeightedVars
##
## Loading required package: Biobase
## Welcome to Bioconductor
##
##      Vignettes contain introductory material; view with
##      'browseVignettes()'. To cite Bioconductor, see
##      'citation("Biobase")', and for packages 'citation("pkgname)".
##

```

```
##
## Attaching package: 'Biobase'
##
## The following object is masked from 'package:MatrixGenerics':
##
##     rowMedians
##
## The following objects are masked from 'package:matrixStats':
##
##     anyMissing, rowMedians
```

```
library(ashr)
```

```
##
## Attaching package: 'ashr'
##
## The following object is masked from 'package:generics':
##
##     prune
```

```
library(pheatmap)
library(PoiClaClu)
library(glmPCA)
```

Project 2 parts 4 and 5 overall: 75 points

Include any code that you used to solve the question as well as the answers for all below questions. In parts 4 and 5 from project 2, you will explore how gene expression differs between electric fish tissues. You will learn how to analyze RNA-seq count data: identify differentially expressed genes, distinguish samples, and make biological interpretations. Additionally, you will visualize findings and communicate them effectively.

Note: Please submit the Rmd and the knit HTML file for this assignment

Dataset reminder:

In this project, you are be analyzing the count data and identifying differentially expressed genes between skeletal muscle and electric organ tissue. As a reminder, these datasets originated from two different electric fish studies (PRJNA1005245 and PRJNA1005244). It may help to consider how these papers analyzed the data: the methods for the PRJNA1005244 dataset are published and the methods for the PRJNA1005245 dataset are written in the third chapter of a thesis. These studies are both using the genus *Campylomormyrus*. **For this assignment, you may not need to analyze all of the data to address the hypothesis.**

Resources:

There are several resources that you may use to complete this assignment: 1) A primary resource: DESeq2 workflow. 2) DESeq2's additional guide 3) Harvard Bioinformatics Core training.

You are expected to use DESeq2 for this assignment. If you decide to use an additional package, please list the package name, how you found it, and why you chose to use it. We will do our best to help you navigate other packages, but we cannot guarantee support for other packages.

Project 2 part 4: Hypothesis testing and exploration

0. In part 4, your goal is to investigate differences in gene expression between the electric organ and skeletal muscle. Write the null hypothesis for this goal below.

There is no significant difference in gene expression between the electric organ and skeletal muscle in the *Campylomormyrus* genus.

1. Load your data into DESeq2

- Your data is located on Talapas: /projects/bgmp/shared/Bi623/Project2/counts.tar
- **Think carefully about your design formula and explain your chosen formula below**
- Note: you can follow the Bear_analysis rmd from class as a template or can also reference other resources online, such as this guide from Duke to load your data.

```
#Include your commands, remember that you might not need to load all of the count files for this hypothesis
campy_meta = read.csv("RNAseq_metadata_campy.csv")
campy_meta = campy_meta %>%
  filter(AgeSize == "adult")

Campy = DESeqDataSetFromHTSeqCount(campy_meta, directory = "counts_tar", design = ~ SpeciesSmaller + Ti
```

```
## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

I chose my design formula because I want to test the effect of tissue (electric organ vs skeletal muscle), controlling for the effect of species.

2. Filter the dataset to remove genes that are lowly expressed

- There are multiple ways to filter RNA-Seq datasets. You can either use cpm as a filter as we did in the Bear_analysis or use raw counts completed in the DESeq2 vignette from the above resources.
- Describe why you used your filtering strategy

```
#Include your commands
campy_cpm = cpm(Campy)
thresh = campy_cpm > 0.05
keep <- rowSums(thresh) >= 2
Campy = Campy[keep,]
```

I used the cpm approach for filtering, since it is the traditional approach.

3. Normalize the dataset and run differential expression analysis

- You may choose to normalize and run differential expression separately as done in the Bear_analysis, or in 1 command as done in the DESeq workflow 5.1.

```
#Include your commands
Campy = DESeq(Campy)
```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

4. Normalize the data to use for dimensionality reduction

- Use the rlog transformation method (follow along with the workflow, section 4.2)
- Describe why the rlog transformation is performed

```
#Include your commands
```

```
campy_rld = rlog(Campy, blind = FALSE)
```

```
## rlog() may take a few minutes with 30 or more samples,
```

```
## vst() is a much faster transformation
```

The rlog transformation stabilizes the variance across the mean, so that methods such as PCA are not dominated by genes with high counts. According to the DESeq workflow, rlog works well on small datasets with a wide range of sequencing depth, which is what we have.

5. Identify distances between samples using sample distance heatmaps

- Generate a Euclidean distance heatmap and dendrogram using your rlog transformed data (follow workflow section 4.3)
- Create a Poisson distance heatmap with dendrogram using DESeq data (follow workflow section 4.3)
- Describe what each of these heatmaps are measuring
- Look at the sample groupings for each heatmap+dendrogram. Do you notice any potential individual outliers? Do both plots agree?

Note: You may need to install the packages “pheatmap” and “RColorBrewer” to follow along. If R tells you to restart your session listen to it.

```
#Include your commands for generating the Euclidean distance heatmap and dendrogram with the rlog data  
sampleDists = dist(t(assay(campy_rld)))
```

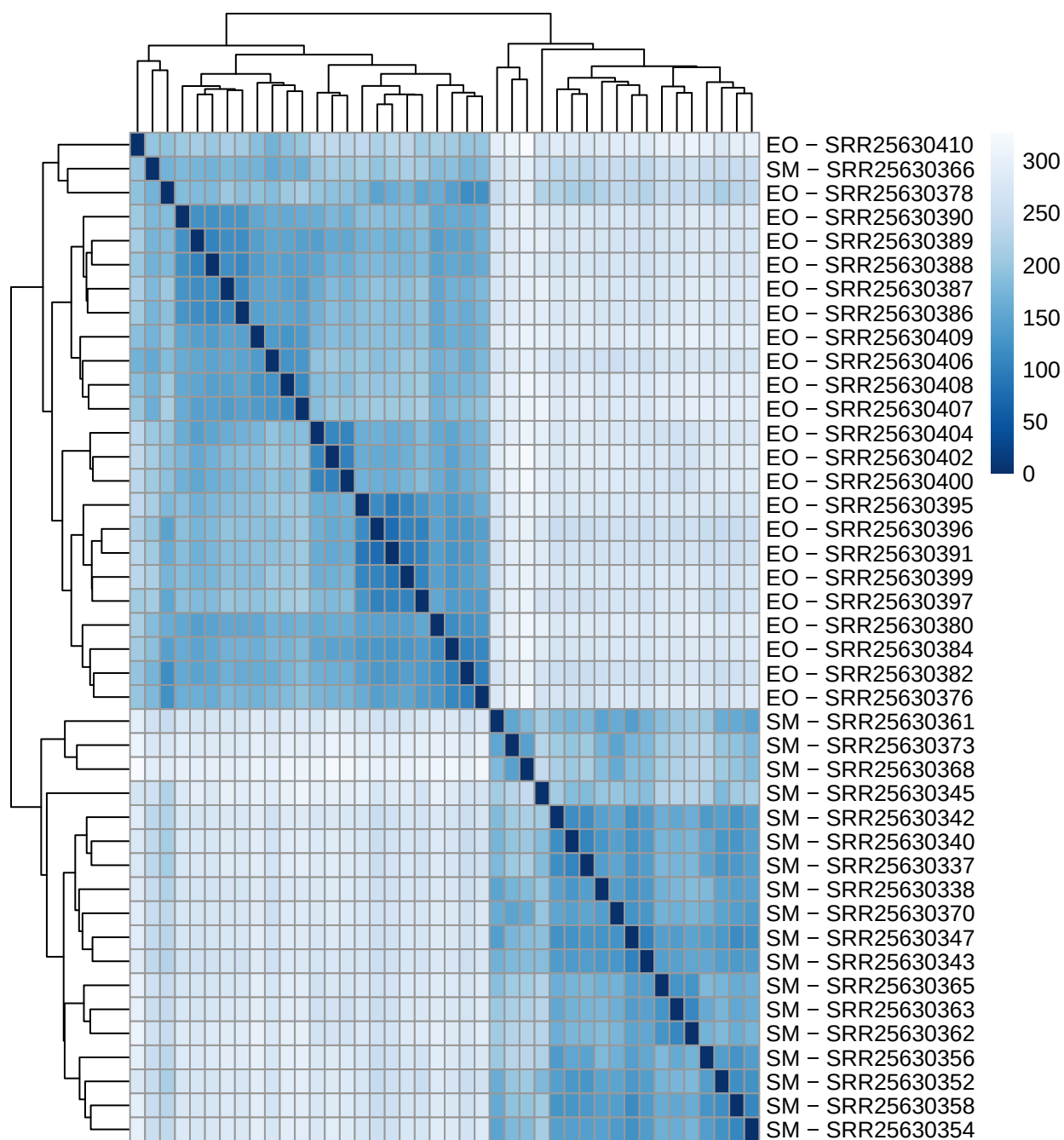
```
sampleDistMatrix = as.matrix(sampleDists)
```

```
rownames(sampleDistMatrix) = paste(campy_rld$Tissue, campy_rld$SRRfile, sep = " - ")
```

```
colnames(sampleDistMatrix) = NULL
```

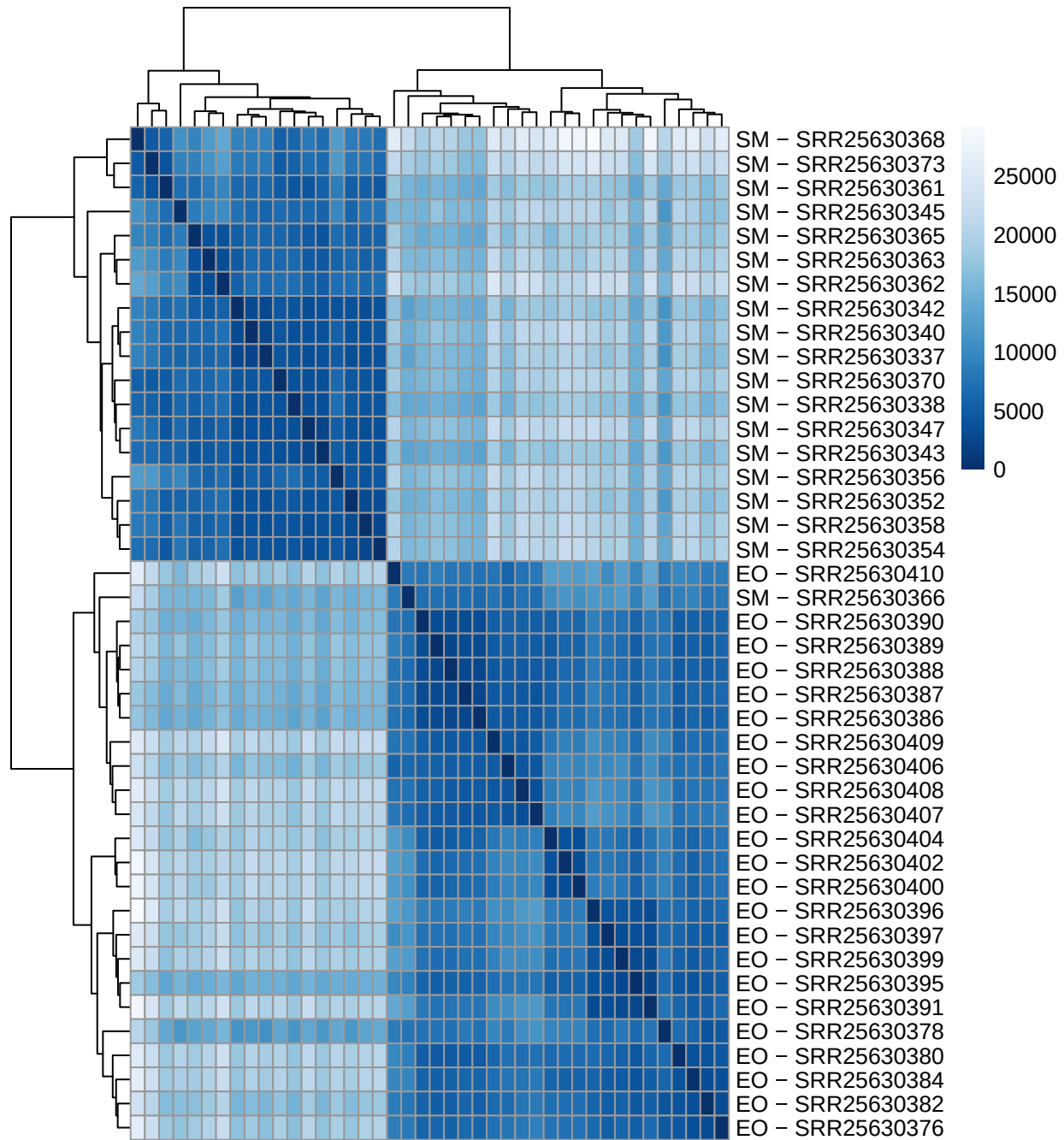
```
colors = colorRampPalette(rev(brewer.pal(9, "Blues")))(255)
```

```
pheatmap(sampleDistMatrix,  
          clustering_distance_rows = sampleDists,  
          clustering_distance_cols = sampleDists,  
          col = colors)
```



```
poisd = PoissonDistance(t(counts(Campy)))

samplePoisDistMatrix <- as.matrix(poisd$dd)
rownames(samplePoisDistMatrix) <- paste(Campy$Tissue, Campy$SRRfile, sep=" - ")
colnames(samplePoisDistMatrix) <- NULL
pheatmap(samplePoisDistMatrix,
          clustering_distance_rows = poisd$dd,
          clustering_distance_cols = poisd$dd,
          col = colors)
```



The two heatmaps are measuring sample-to-sample distance. The first is using Euclidean distance and normalized data, and the second is using Poisson distance and unnormalized data. Both plots seem to suggest that SRR25630366 may be an outlier, since it is not grouped with the correct tissue.

6. Examine similarity in sample global gene expression using PCA, Generalized PCA, and MDS plots.

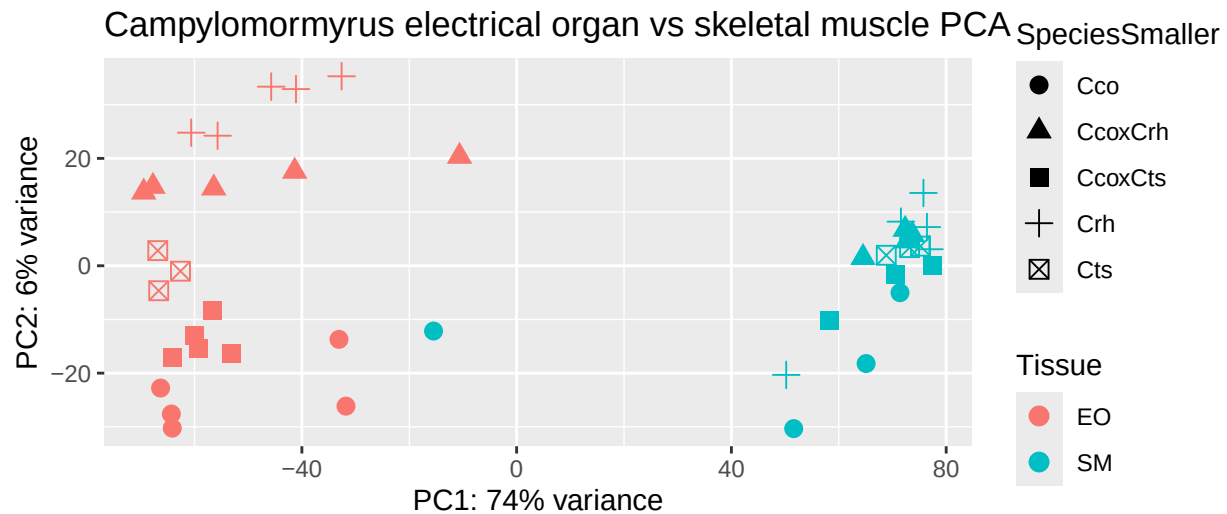
- Generate PCA, Generalized PCA, and MDS plots (following workflow section 4.4-4.6)
- Modify the color and shape of points to reflect Tissue and Species of samples
- Briefly describe each approach and their findings.

```
pcaData = plotPCA(campy_rld, intgroup = c("Tissue", "SpeciesSmaller"), returnData = TRUE)
```

```
## using ntop=500 top features by variance
```

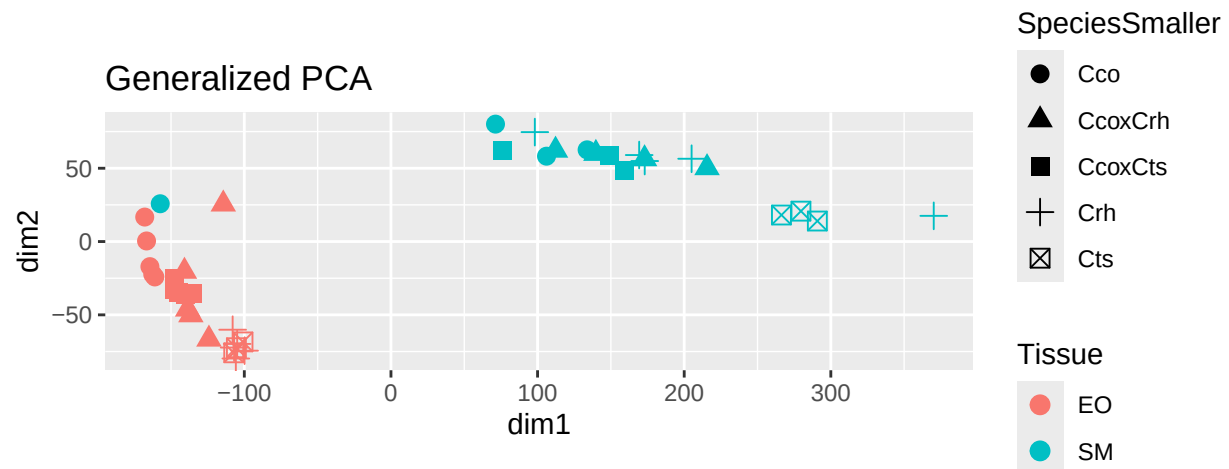
```
percentVar = round(100 * attr(pcaData, "percentVar"))

ggplot(pcaData, aes(x = PC1, y = PC2, color = Tissue, shape = SpeciesSmaller)) +
  geom_point(size = 3) +
  xlab(paste0("PC1: ", percentVar[1], "% variance")) +
  ylab(paste0("PC2: ", percentVar[2], "% variance")) +
  coord_fixed() +
  ggtitle("Campylomormyrus electrical organ vs skeletal muscle PCA")
```

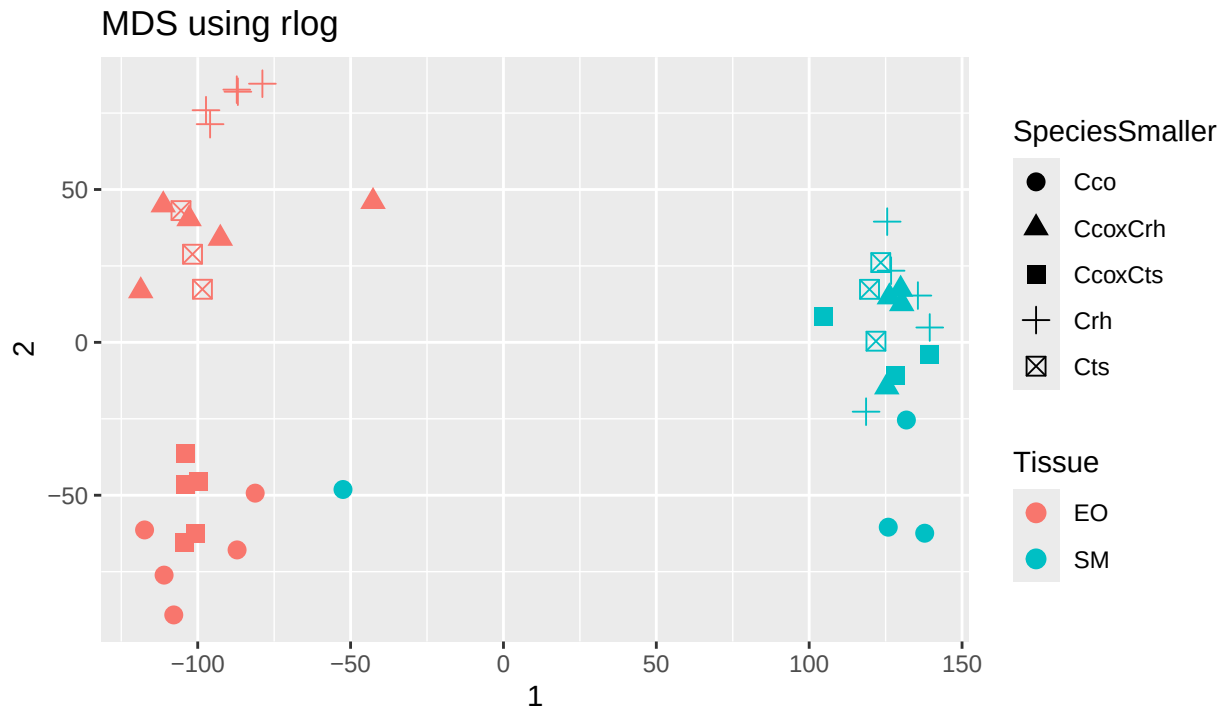


```
gpca = glmpca(counts(Campy), L = 2)
gpca.dat = gpca$factors
gpca.dat$Tissue = Campy$Tissue
gpca.dat$SpeciesSmaller = Campy$SpeciesSmaller

ggplot(gpca.dat, aes(x = dim1, y = dim2, color = Tissue, shape = SpeciesSmaller)) +
  geom_point(size = 3) +
  coord_fixed() +
  ggtitle("Generalized PCA")
```



```
mds = as.data.frame(colData(campy_rld)) %>%
  cbind(cmdscale(sampleDistMatrix))
ggplot(mds, aes(x = `1`, y = `2`, color = Tissue, shape = SpeciesSmaller)) +
  geom_point(size = 3) +
  coord_fixed() +
  ggtitle("MDS using rlog")
```



These figures appear to reject the null hypothesis, since there is a clear distinction between the two tissue types, showing evidence that there is a difference in gene expression between the electric organ and skeletal muscle in the *Campylomormyrus* genus. Both methods (PCA and MDS) identify the same outlier. Moreover, PCA does not use normalized data and captures variance better, whereas MDS uses normalized data and preserves pairwise distances.

7. Use your dimensionality reduction plots, evaluate sample quality and complete quality adjustments.

- Based on your dimensionality reduction plots, do you have concerns about any samples? Explain.
- If so, remove them now, repeat steps of loading in data, filtering, normalization, and running the differential expression analysis.

It seems like one of the samples was mislabeled (i.e., was an EO tissue sample but was labeled as SM), since it clustered with the rest of the EO samples in all three plots but is supposedly an SM sample. This sample also shows up in the heat maps and has SRR file SRR25630366.

```
campy_meta_f = campy_meta %>%
  filter(SRRfile != "SRR25630366")

Campy_f = DESeqDataSetFromHTSeqCount(campy_meta_f, directory = "counts_tar", design = ~ SpeciesSmaller + Tissue)

## Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in
## design formula are characters, converting to factors
```

```

campy_cpm_f = cpm(Campy_f)
thresh_f = campy_cpm_f > 0.05
keep_f <- rowSums(thresh_f) >= 2
Campy_f = Campy_f[keep_f,]

Campy_f = DESeq(Campy_f)

```

```
## estimating size factors
```

```
## estimating dispersions
```

```
## gene-wise dispersion estimates
```

```
## mean-dispersion relationship
```

```
## final dispersion estimates
```

```
## fitting model and testing
```

8. Identify genes that are differentially expressed between SM and EO tissues in EVERY species comparison.

- Generate a results table using log fold shrinkage and proper contrasts
- Refer to this reference for guidance on extracting contrasts
- When setting contrasts, identify the baseline variables used in the intercept to help determine proper contrasts
- Filter DEGs to contain only candidates with a $|\log_2\text{FoldChange}| > 1$ and adjusted value $< .05$
- Join DEG data sets for every species, remove any DEG not found in every species, and calculate the average log2FC and average adjusted p-value across all species
- Visualize your cleaned DEG dataset with a volcano plot.
 - Use the average log2FC and adjusted p-values for plotting
 - Make sure your plot indicates the meaning of negative and positive logfold changes

```
resultsNames(Campy_f)
```

```
## [1] "Intercept" "SpeciesSmaller_CcoxCrh_vs_Cco"
## [3] "SpeciesSmaller_CcoxCts_vs_Cco" "SpeciesSmaller_Crh_vs_Cco"
## [5] "SpeciesSmaller_Cts_vs_Cco" "Tissue_SM_vs_EO"
## [7] "SpeciesSmallerCcoxCrh.TissueSM" "SpeciesSmallerCcoxCts.TissueSM"
## [9] "SpeciesSmallerCrh.TissueSM" "SpeciesSmallerCts.TissueSM"

```

```

Campy_Cco_SMvsEO_results = data.frame(lfcShrink(
  Campy_f,
  coef = "Tissue_SM_vs_EO",
  type = "ashr"
)) %>%
rownames_to_column(var = "genes") %>%
filter(padj < 0.05) %>%
filter(abs(log2FoldChange) > 1) %>%

```

```
dplyr::rename(
  "Cco_log2FoldChange" = "log2FoldChange",
  "Cco_pvalue_adj" = "padj"
) %>%
dplyr::select(genes, Cco_log2FoldChange, Cco_pvalue_adj)
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
##     Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
##     https://doi.org/10.1093/biostatistics/kxw041
```

```
Campy_Crh_SMvsEO_results = data.frame(lfcShrink(
  Campy_f,
  type = "ashr",
  contrast = list(c("Tissue_SM_vs_EO", "SpeciesSmallerCrh.TissueSM"))
)) %>%
rownames_to_column(var = "genes") %>%
filter(padj < 0.05) %>%
filter(abs(log2FoldChange) > 1) %>%
dplyr::rename(
  "Crh_log2FoldChange" = "log2FoldChange",
  "Crh_pvalue_adj" = "padj"
) %>%
dplyr::select(genes, Crh_log2FoldChange, Crh_pvalue_adj)
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
##     Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
##     https://doi.org/10.1093/biostatistics/kxw041
```

```
Campy_Cto_SMvsEO_results = data.frame(lfcShrink(
  Campy_f,
  type = "ashr",
  contrast = list(c("Tissue_SM_vs_EO", "SpeciesSmallerCts.TissueSM"))
)) %>%
rownames_to_column(var = "genes") %>%
filter(padj < 0.05) %>%
filter(abs(log2FoldChange) > 1) %>%
dplyr::rename(
  "Cts_log2FoldChange" = "log2FoldChange",
  "Cts_pvalue_adj" = "padj"
) %>%
dplyr::select(genes, Cts_log2FoldChange, Cts_pvalue_adj)
```

```
## using 'ashr' for LFC shrinkage. If used in published research, please cite:
##     Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
##     https://doi.org/10.1093/biostatistics/kxw041
```

```
Campy_CcoxCts_SMvsEO_results = data.frame(lfcShrink(
  Campy_f,
  type = "ashr",
  contrast = list(c("Tissue_SM_vs_EO", "SpeciesSmallerCcoxCts.TissueSM"))
)) %>%
```

```

rownames_to_column(var = "genes") %>%
filter(padj < 0.05) %>%
filter(abs(log2FoldChange) > 1) %>%
dplyr::rename(
  "CcoxCts_log2FoldChange" = "log2FoldChange",
  "CcoxCts_pvalue_adj" = "padj"
) %>%
dplyr::select(genes, CcoxCts_log2FoldChange, CcoxCts_pvalue_adj)

```

```

## using 'ashr' for LFC shrinkage. If used in published research, please cite:
##     Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
##     https://doi.org/10.1093/biostatistics/kxw041

```

```

Campy_CcoxCrh_SMvsEO_results = data.frame(lfcShrink(
  Campy_f,
  type = "ashr",
  contrast = list(c("Tissue_SM_vs_EO", "SpeciesSmallerCcoxCrh.TissueSM")))
) %>%
rownames_to_column(var = "genes") %>%
filter(padj < 0.05) %>%
filter(abs(log2FoldChange) > 1) %>%
dplyr::rename(
  "CcoxCrh_log2FoldChange" = "log2FoldChange",
  "CcoxCrh_pvalue_adj" = "padj"
) %>%
dplyr::select(genes, CcoxCrh_log2FoldChange, CcoxCrh_pvalue_adj)

```

```

## using 'ashr' for LFC shrinkage. If used in published research, please cite:
##     Stephens, M. (2016) False discovery rates: a new deal. Biostatistics, 18:2.
##     https://doi.org/10.1093/biostatistics/kxw041

```

```

joined_results = Campy_Cco_SMvsEO_results %>%
  inner_join(Campy_Cto_SMvsEO_results) %>%
  inner_join(Campy_Crh_SMvsEO_results) %>%
  inner_join(Campy_CcoxCrh_SMvsEO_results) %>%
  inner_join(Campy_CcoxCts_SMvsEO_results)

```

```

## Joining with 'by = join_by(genes)'
## Joining with 'by = join_by(genes)'
## Joining with 'by = join_by(genes)'
## Joining with 'by = join_by(genes)'

```

```

campy_results = joined_results %>%
  group_by(genes) %>%
  summarize(
    avg_log2FC = mean(c_across(ends_with("log2FoldChange"))),
    avg_pval = mean(c_across(ends_with("pvalue_adj")))
  )

alpha = 0.01
campy_results$sig = -log10(campy_results$avg_pval)

```

```

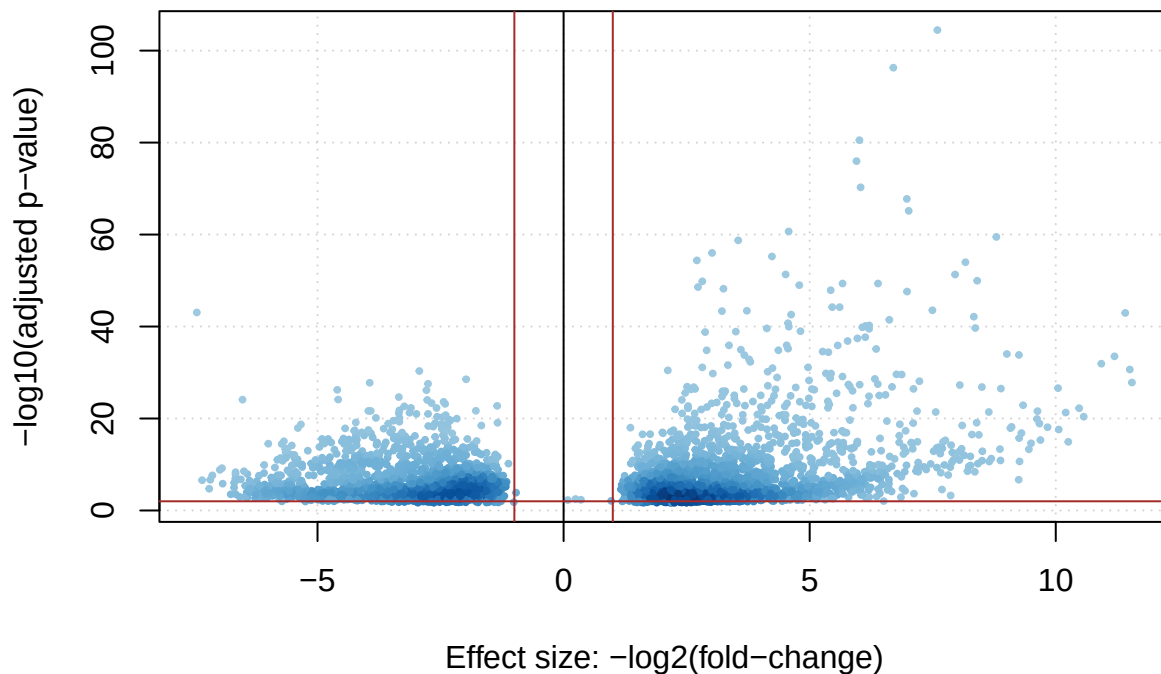
campy_results[is.infinite(campy_results$avg_log2FC), "sig"] = 350

genes.to.plot = !is.na(campy_results$avg_pval)

cols <- densCols(campy_results$avg_log2FC, campy_results$avg_pval)
cols[campy_results$avg_pval == 0] <- "purple"
campy_results$pch <- 19
campy_results$pch[campy_results$avg_pval == 0] <- 6
plot(-campy_results$avg_log2FC,
     campy_results$avg_pval,
     col=cols, panel.first=grid(),
     main="Volcano plot (right is up in EO, left up in SM)",
     xlab="Effect size: -log2(fold-change)",
     ylab="-log10(adjusted p-value)",
     pch=campy_results$pch, cex=0.4)
abline(v=0)
abline(v=c(-1,1), col="brown")
abline(h=-log10(alpha), col="brown")

```

Volcano plot (right is up in EO, left up in SM)



9. Visualize individual candidates

- Using the DEG dataframe from above, report out the top five candidates with the lowest average adjusted p-value.

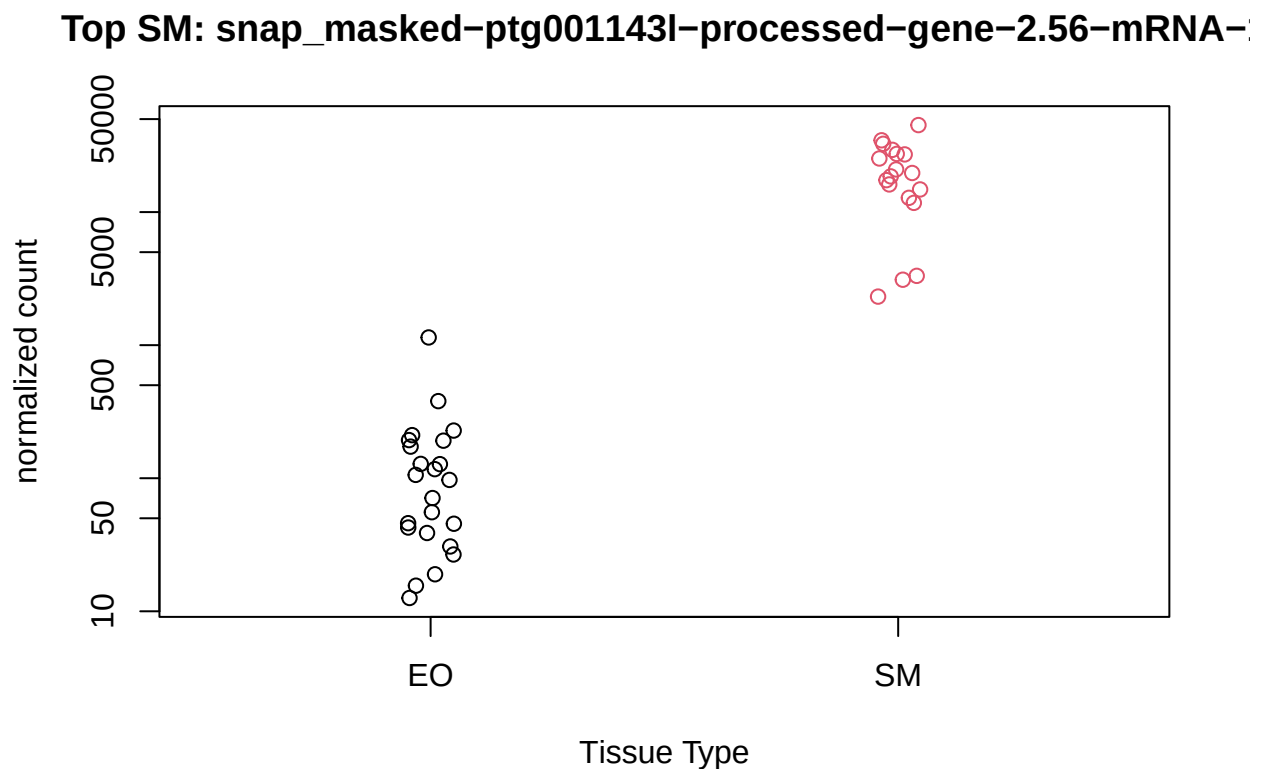
The top five candidates:

1. maker-ptg000555l-snap-gene-1.19-mRNA-1
2. maker-ptg001270l-snap-gene-47.16-mRNA-1
3. maker-ptg000070l-snap-gene-5.18-mRNA-1
4. snap_masked-ptg000102l-processed-gene-12.45-mRNA-1
5. maker-ptg000361l-snap-gene-40.32-mRNA-1

- Using the DEG dataframe from above, visualize the normalized counts for the gene with the highest log2FC in the SM. Also, make a plot for the top candidate for the EO.

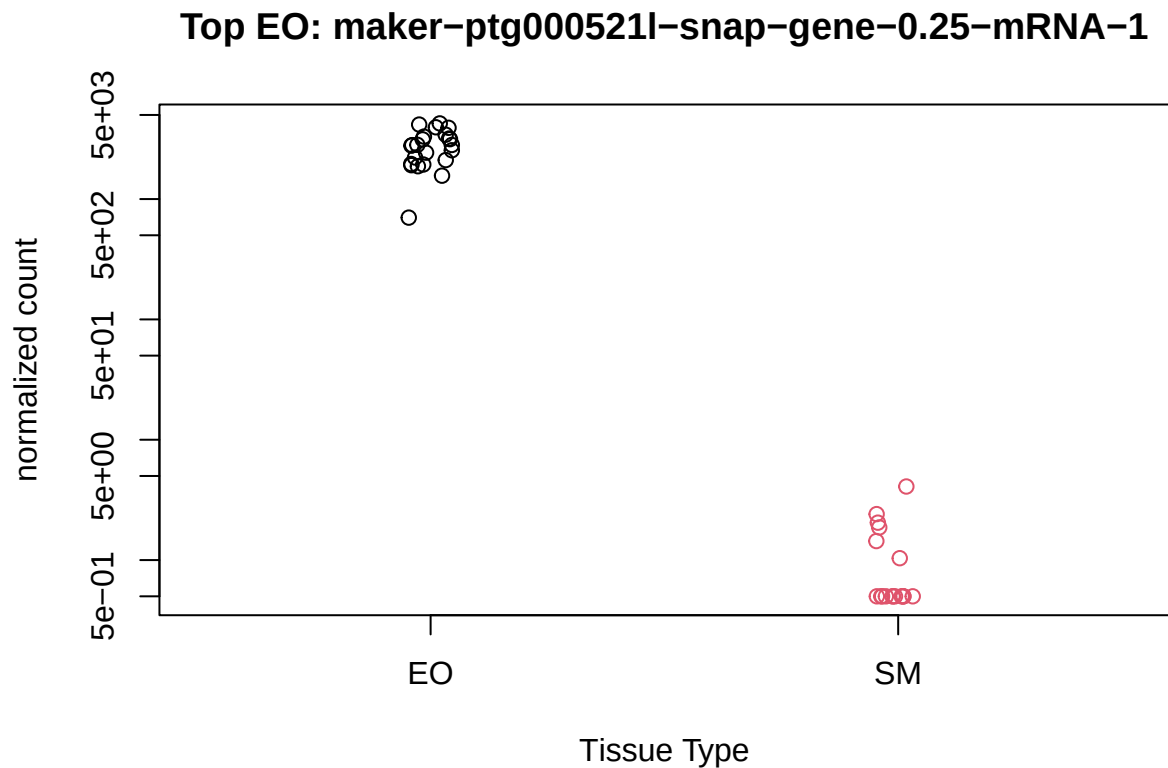
```
# Top candidate for SM: snap_masked-ptg001143l-processed-gene-2.56-mRNA-1
# Top candidate for EO: maker-ptg000521l-snap-gene-0.25-mRNA-1
```

```
plotCounts(
  Campy_f,
  "snap_masked-ptg001143l-processed-gene-2.56-mRNA-1",
  intgroup = "Tissue",
  normalized = TRUE,
  col = Campy_f$Tissue,
  xlab = "Tissue Type",
  main = "Top SM: snap_masked-ptg001143l-processed-gene-2.56-mRNA-1"
)
```



```
plotCounts(
  Campy_f,
  "maker-ptg000521l-snap-gene-0.25-mRNA-1",
  intgroup = "Tissue",
```

```
normalized = TRUE,
col = Campy_f$Tissue,
xlab = "Tissue Type",
main = "Top EO: maker-ptg0005211-snap-gene-0.25-mRNA-1"
)
```



10. Using the DESeq LRT test, evaluate whether there is an association between gene expression and electric discharge signal duration
 - Complete a LRT test using design formulas to determine whether there is an interaction between duration and electric discharge signal
 - You may need to reload in the data to DESeq2 and re-process it depending on how you specified your design formula.
 - Follow this resource to guide your approach
 - Use only the purebred species for this analysis (NO hybrids, i.e., do not include ccoxcts or ccoxrh in the analysis)
 - Use degPatterns to sort genes into groups that correspond to EOD duration
 - From the output, identify groups that 1) have higher gene expression in the EO than SM and 2) have gene expression patterns that increase or decrease with EOD duration
 - Create a plot from degPlotCluster to showcase ONLY these selected groups (ex: a group of genes that has a higher abundance in longer compared to shorter duration electric signals)

```
## YOUR CODE HERE
```

11. Based on the signal duration clustering, select 1 of the relevant clusters and complete a gene ontology analysis.

- Isolate a gene list for the cluster of interest
- Convert Campy gene names to D. rerio gene names using the orthology table derived from our ICA2 Genespace assignment (Danio_Campylomormyrus_1to1_orthologs_table.csv)
 - Note: degPatterns transforms the Campy gene names, make sure to change them back so they match with the orthology table
- Complete a gene ontology analysis on the cluster of interest using the zebrafish gene names
 - Note: enrichGO and enrichKEGG are options, use this reference

```
#Include commands
```

Project 2 Part 5: Creating high quality figures for your report

Go back through your work from part 1-4 and identify the key pieces of information and plots for writing your project 2 report. Remake your chosen plots below, creating publication quality figures. Your project 2 report should have 4-6 figures, these can have multiple panels if you know how to achieve that in R. Each figure should add a different piece of information (i.e., do NOT include 3 dimensionality plots that show the same pattern).

Guidelines:

- Figures should have a title and axes that are easy to read
- Have an accessible color scheme
- Have scientifically relevant information highlighted (ex: color scheme denotes different species)
- When applicable, color schemes should be consistent across plots

1. Figure 1

```
#Make figure
```

```
#OR delete this box and link a figure that you made outside of R
```

2. Figure 2

```
#Make figure
```

```
#OR delete this box and link a figure that you made outside of R
```

3. Figure 3

```
#Make figure
```

```
#OR delete this box and link a figure that you made outside of R
```

4. Figure 4

```
#Make figure  
#OR delete this box and link a figure that you made outside of R
```

5. Figure 5

```
#OPTIONAL analysis
```

5. Figure 6

```
#OPTIONAL analysis
```